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Population-specific distribution of transcription factor 7-like 2 rs12255372 in Indonesian adults with and without Type 2 diabetes

Running title: TCF7L2 rs12255372 in Indonesian adults

Prince Saputra[1], Miswar Fattah[2], Himawan Sanusi[3,4], Andi Alfian Zainuddin[5], Rusdina Bte Ladju[6], Mansyur Arif[7] and Liong Boy Kurniawan[7]*

[1] Doctoral Program, Faculty of Medicine, Hasanuddin University, Makassar 90245, Indonesia

[2] Research and Specialty Test Development Division, PT Prodia Widyahusada Tbk, Jakarta 10430, Indonesia

[3] Department of Internal Medicine, Faculty of Medicine, Hasanuddin University, Makassar, Indonesia

[4] Department of Community Medicine, Faculty of Medicine, Hasanuddin University, Makassar 90245, Indonesia

[5] Department of Pharmacology, Faculty of Medicine, Hasanuddin University, Makassar 90245, Indonesia

[6] Department of Clinical Pathology, Faculty of Medicine, Hasanuddin University, Makassar 90245, Indonesia

[7] Dr. Wahidin Sudirohusodo National General Hospital, Hasanuddin University, Makassar, Indonesia

Corresponding author: Liong Boy Kurniawan, Department of Clinical Pathology, Faculty of Medicine, Hasanuddin University, Perintis Kemerdekaan Street Km. 10, Tamalanrea, Makassar 90245, South Sulawesi, Indonesia; Email: liongboykurniawan@yahoo.com

ORCID iDs: Prince Saputra, 0009-0004-2891-7002; Miswar Fattah, 0000-0003-3679-5901; Himawan Sanusi, 0000-0003-3828-2958; Andi Alfian Zainuddin, 0000-0003-4436-3906; Rusdina Bte Ladju, 0000-0001-5393-0429; Mansyur Arif, 0000-0002-9104-8127; Liong Boy Kurniawan, 0000-0002-2633-0441

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Conflicts of interest

The authors declare that they have no conflicts of interest.

Novelty statement

- **What is already known?** Repeated studies have linked common TCF7L2 variants to Type 2 diabetes. However, rs12255372 differs markedly across populations in both frequency and effect, and little evidence is available from Indonesia.
- **What this study has found?** The rs12255372 T allele was rare in this Indonesian cohort, no TT genotype was observed, and there was no detectable association with Type 2 diabetes, C-peptide or HOMA-IR.
- **What are the implications?** Diabetes-associated variants identified in other populations should be locally validated before use in risk stratification in Indonesia.

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Abstract

Aims: To characterise rs12255372 in TCF7L2 (transcription factor 7-like 2) among Indonesian adults and assess its relationship to Type 2 diabetes, C-peptide concentrations and insulin resistance.

Methods: The case-control sample comprised 111 adults with Type 2 diabetes and 60 non-diabetic controls; another 10 adults with Type 1 diabetes provided descriptive metabolic comparisons only, giving 181 participants overall. Genotypes were determined by polymerase chain reaction followed by Sanger sequencing. Biochemical assessment covered fasting glucose and insulin, glycated haemoglobin, C-peptide and an insulin resistance estimate derived from homeostasis model assessment (HOMA-IR). Cases and controls formed the primary genetic comparison, with age, body mass index and sex incorporated into a multivariable logistic model.

Results: GG accounted for 96% of cases and 97% of controls, while the corresponding GT proportions were 3.6% and 3.3%; TT was absent. The case-control comparison yielded no association for rs12255372 (odds ratio 1.08, 95% CI 0.19, 6.16; P = 0.930). Neither C-peptide (P = 0.157) nor HOMA-IR (P = 0.609) showed a significant genotype difference, and adjustment did not identify a significant genotype effect.

Conclusions: The rs12255372 T allele was uncommon in this Indonesian cohort, and no detectable association with Type 2 diabetes or the assessed metabolic phenotypes was identified. Testing whether diabetes-related genetic markers apply across Indonesia will require larger samples recruited at multiple centres.

Keywords: Transcription Factor 7-Like 2 Protein; Diabetes Mellitus, Type 2; C-Peptide; Insulin Resistance; Genetic Association Studies; Indonesia

Introduction

Long-term disruption of glucose metabolism in Type 2 diabetes reflects inadequate insulin release and diminished insulin responsiveness, together with environmental influences and inherited susceptibility. The disease imposes substantial healthcare costs and contributes to disability and deaths occurring prematurely. Worldwide disease burden is still increasing, with Indonesia and other countries in the low- and middle-income groups experiencing especially rapid growth.[1-4]

Because genetic architecture differs between populations, locally generated evidence is needed before risk variants identified elsewhere are used for prediction or stratification.

Genetic studies repeatedly implicate TCF7L2, which encodes transcription factor 7-like 2, placing it among the best-replicated susceptibility loci for Type 2 diabetes. At chromosome 10q25.2-q25.3, this gene encodes a regulator of transcription that participates in the Wnt/beta-catenin pathway, responses to incretins, beta-cell activity in the pancreas, insulin release and the maintenance of glucose balance.[5-8] Several populations show links between the intronic G>T variant rs12255372, Type 2 diabetes and changes in beta-cell function. Its allele frequency and estimated effects nevertheless differ markedly with population background, patterns of linkage disequilibrium and environmental conditions.[9-15]

Evidence from Indonesia remains limited, particularly studies that combine rs12255372 genotyping with metabolic indicators of endogenous insulin secretion and insulin resistance. Endogenous insulin production is reflected by C-peptide, while resistance to insulin can be estimated through homeostasis model assessment (HOMA-IR).[16-18] We investigated how rs12255372 was distributed among Indonesian adults and whether it related to diabetes case status or to C-peptide and HOMA-IR measurements. A small Type 1 diabetes group was retained only as a descriptive metabolic comparison group to illustrate severe beta-cell dysfunction.

Methods

Ethics and study design

Approval was granted by the Health Research Ethics Committee of Hasanuddin University, Indonesia, under reference 563/UN4.6.4.5.31/PP36/2025, and study procedures followed the Declaration of Helsinki. Every participant provided informed consent in writing before entering the study.

An observational case-control design was used during the July–December 2025 study period. Participants were recruited and blood samples were collected in Samarinda, East Kalimantan, Indonesia. Recruitment took place at healthcare facilities and research laboratories specified in the study protocol. Selection followed the inclusion requirements and exclusions established before recruitment. Laboratory examinations were performed at the Biomolecular Teaching Hospital Laboratory of HUM-RC, Hasanuddin University, Makassar, South Sulawesi, Indonesia. The study

size reflected the total number of eligible participants recruited during July–December 2025 for whom the required study data were available. A total of 181 adults were enrolled: 111 with Type 2 diabetes, 60 controls without diabetes and 10 with Type 1 diabetes included only as a descriptive metabolic comparison group. The enrolled adults ranged from 18 to 49 years of age and comprised 29 men and 152 women.

Case identification followed the 2025 diabetes diagnostic guidance in the American Diabetes Association Standards of Care in Diabetes.[19] For controls, fasting glycaemia was compatible with a non-diabetic state, and no prior diabetes diagnosis was reported. Participants with Type 1 diabetes were identified by a physician on the basis of the clinical diagnosis documented in the medical record, with age considered as part of the clinical assessment. Eligibility required an age of 18-49 years and written informed consent, and recruitment was open to both men and women. Individuals with severe systemic illness, pregnancy, malignancy or acute infection were excluded. Participants with insufficient data for a particular analysis were excluded from that analysis.

Clinical and biochemical measurements

Following an overnight period without food, blood samples were drawn from a peripheral vein. An enzymatic assay on the BIOLIS 50i Superior automated clinical analyser (Biolis Medical Corp., Hino, Tokyo, Japan) quantified glucose in fasting samples. Glycated haemoglobin (HbA1c) was measured using an ADAMS HA-8380V glycohaemoglobin analyser (ARKRAY Factory, Inc., Koka, Shiga, Japan) based on ion-exchange chromatography. Fasting insulin and C-peptide concentrations were measured using a VIRTUEDX VERCENTRA CS-1500 analyser (PT Darma Labora Indonesia, Cikarang, Bekasi, West Java, Indonesia) with a chemiluminescence immunoassay method. To derive HOMA-IR, the product of fasting insulin in mIU/L and fasting glucose in mmol/L was divided by 22.5. C-peptide was used as an indicator of endogenous insulin secretion.

DNA extraction, genotyping and sequence analysis

EDTA tubes were used to receive 3-5 mL samples drawn from a peripheral vein. For DNA isolation, 200 µL aliquots of whole blood were processed with the Geneaid Genomic DNA Mini Kit (Blood/Cultured Cell; Cat. No. GB100; Geneaid Biotech Ltd., New Taipei City, Taiwan), following the protocol supplied with the kit. The procedure included proteinase K digestion, cell

lysis, binding to a silica membrane, washing and elution in 100 µL of elution buffer. Purified DNA was stored at -40 °C until amplification.

Amplification targeted a 346-bp segment spanning TCF7L2 rs12255372, with 5'-CTGGAAACTAAGGCGTGAGG-3' as the forward primer and 5'-GGGTCGATGTTGTTGAGCTT-3' as the reverse primer. The reactions used a 2x PCR Master Mix with Taq DNA polymerase and were run on a T100 Thermal Cycler (Bio-Rad Laboratories, Inc., Hercules, CA, USA). The programme began with 5 min at 95 °C. Each of the subsequent 40 cycles consisted of 15 s at 95 °C for denaturation, 15 s at 65 °C for annealing and 45 s at 72 °C for extension. A further 15 min at 72 °C completed the extension step. 1st BASE Sanger Sequencing Services sequenced the amplified products from all 181 samples on an ABI PRISM 3730xl Genetic Analyzer (Applied Biosystems, Foster City, CA, USA).

Geneious Prime 2025.2 was used to examine the sequencing traces, and the variant assignment was checked against the rs12255372 record in NCBI dbSNP. The locus was mapped to the GRCh38.p14 human genome assembly and is described in HGVS genomic nomenclature as NC_000010.11:g.113049143G>T. A single guanine peak at the polymorphic locus was classified as GG, whereas overlapping guanine and thymine peaks were classified as GT. No TT genotype was detected.

Statistical analysis

Each analysis was restricted to eligible participants whose records contained the information needed for the variables under examination. The enrolled sample comprised 181 participants; the number included in an individual analysis depended on the availability of the relevant data or genotype. Depending on their distribution, continuous measurements are summarised by the mean with its standard deviation or by the median with its interquartile range. Frequencies describe categorical measurements. Comparisons across the three study groups used one-way analysis of variance when continuous measurements followed a normal distribution; otherwise, the Kruskal-Wallis procedure was applied. The comparison of sex frequencies used Fisher's exact procedure.

For the primary genetic comparison, Fisher's exact procedure assessed differences in genotype and allele frequencies between cases and controls. Control genotypes were also examined for consistency with Hardy-Weinberg expectations. Within the genotyped cohort, exploratory

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comparisons of C-peptide and HOMA-IR by genotype used the Mann-Whitney U procedure, with each comparison restricted to participants having the necessary measurements.

A multivariable logistic model assessed which factors related independently to being a Type 2 diabetes case. Diabetes status was coded as 0 for controls and 1 for Type 2 diabetes; rs12255372 genotype as 0 for GG and 1 for GT; and sex as 0 for women and 1 for men. Age, body mass index (BMI) and sex entered the model as covariates when estimating the relationship between GT carriage and diabetes case status. Estimates are expressed as odds ratios (ORs), accompanied by 95% confidence intervals (CIs). A P value below 0.05 was the threshold for statistical significance. All statistical calculations used version 26.0 of IBM SPSS Statistics (IBM Corp., Armonk, NY, USA). Because the rs12255372 T allele was rare and no TT homozygotes were identified, power to detect modest genetic effects was limited. The manuscript followed the STREGA recommendations (Strengthening the Reporting of Genetic Association Studies) for reporting genetic association research.[20]

Results

Participant characteristics

Table 1 presents the clinical and biochemical information available for the 181 enrolled adults. Relative to controls, the Type 2 diabetes group had a higher age and higher measurements of fasting glucose, HbA1c, insulin and HOMA-IR. The Type 1 diabetes descriptive group had markedly lower C-peptide concentrations, consistent with severe loss of endogenous insulin secretion.

Genotype distribution and association with Type 2 diabetes

The GG genotype predominated in both the Type 2 diabetes and control groups. GT was detected in 4 of 111 participants with Type 2 diabetes (3.6%) and 2 of 60 controls (3.3%); no TT genotype was observed. The control genotype distribution did not deviate from Hardy-Weinberg equilibrium (P = 1.000). Neither genotype nor allele distribution differed significantly between groups. Using GG as the reference, the GT comparison provided no evidence of an association with diabetes case status (OR 1.08, 95% CI 0.19, 6.16; P = 0.930; Table 2). Representative chromatograms are shown in Figure 1.

C-peptide and HOMA-IR by genotype

Exploratory genotype-phenotype analyses were conducted within the genotyped cohort, including the Type 1 diabetes descriptive group, using participants with the data required for each outcome. Median C-peptide concentrations were 0.79 (0.70, 0.86) nmol/L in GT carriers and 0.70 (0.50, 0.83) nmol/L in GG carriers (P = 0.157). Median HOMA-IR was 3.0 (1.9, 3.7) in GT carriers and 2.7 (1.6, 3.6) in GG carriers (P = 0.609) (Table 3). Because only six GT carriers were identified and the Type 1 diabetes group was included only for descriptive comparison, these exploratory findings should be interpreted cautiously.

Multivariable analysis

After adjustment in the logistic model, GT carriage at rs12255372 showed no independent relationship to diabetes case status (adjusted OR 0.646, 95% CI 0.070, 5.969; P = 0.700). The estimated odds of case status rose with age (adjusted OR 1.323, 95% CI 1.199, 1.459; P < 0.001) and declined with BMI (adjusted OR 0.752, 95% CI 0.677, 0.835; P < 0.001). The male-to-female comparison indicated greater odds among men (adjusted OR 3.077, 95% CI 1.020, 9.283; P = 0.046; Table 4).

Discussion

The rs12255372 T allele occurred only rarely among the Indonesian adults examined, which was the central observation of this study. Only six heterozygous participants were identified, no TT homozygotes were observed, and no detectable association with Type 2 diabetes was found. Comparisons by genotype likewise revealed no significant difference in either C-peptide or HOMA-IR. With so few minor-allele carriers, this marker offered little discrimination within the sample. The findings therefore do not establish that TCF7L2 lacks a role in Type 2 diabetes.

In studies from Europe, North Africa, the Middle East and South Asia, the T allele has generally been reported more frequently and linked to Type 2 diabetes more often than in the present cohort.[9-15] This contrast supports the population dependence of genetic susceptibility signals. The cohort included participants from Banjar, Dayak, Javanese, Bugis, Butonese and Minangkabau backgrounds, illustrating substantial population diversity within Indonesia. Differences in allele frequency, linkage disequilibrium and haplotype structure can alter the measurable effect of a

single marker. Accordingly, a risk variant identified in one population should not automatically be assumed to have the same predictive value in another.

The biological functions of TCF7L2 include regulation of beta-cell activity in the pancreas, responses to incretins, insulin release and maintenance of glucose balance.[5,8,16-18] The concurrent assessment of C-peptide and HOMA-IR was intended to explore whether rs12255372 tracked with endogenous insulin secretion or insulin resistance. Only six participants carried GT, which limits the interpretation of the non-significant genotype comparisons. The wide confidence interval around the Type 2 diabetes association likewise indicates substantial uncertainty rather than evidence of equivalence.

Genotype did not show an independent relationship to diabetes case status after covariate adjustment. Case status was positively related to age and sex but negatively related to BMI. Because cases and controls were not matched for age or sex, residual confounding is possible despite multivariable adjustment. These clinical associations therefore should not be interpreted causally and were included primarily to account for group imbalance in the genetic model.

Strengths of the study include Sanger sequencing-based genotype classification and concurrent evaluation of biochemical measures of glucose metabolism. The study also adds data from Indonesia, a population that remains under-represented in global diabetes genetics. Recent research increasingly emphasises ancestry-aware and multi-population approaches to risk architecture and functional interpretation of non-coding variants.[21-27] The present findings reinforce the need to validate candidate genetic markers locally before incorporating them into risk prediction or stratification.

Several limitations should be considered. The minor allele was rare, no TT homozygotes were identified and the number of GT carriers was insufficient for precise genotype-stratified estimates. Only one TCF7L2 variant was examined, preventing haplotype or multi-variant analysis. Recruitment was limited to a single Indonesian region and the modest sample size limits generalisability across the country. The exploratory genotype-phenotype analyses also included the small Type 1 diabetes descriptive group, which may influence C-peptide and HOMA-IR comparisons. Finally, the study did not include ancestry-informative markers, so the observed population differences cannot be attributed to genetic ancestry directly.

In conclusion, rs12255372 was uncommon in this Indonesian cohort and showed no detectable association with Type 2 diabetes, C-peptide or HOMA-IR. The findings highlight the importance of population-specific validation of diabetes-associated variants and support larger multicentre studies incorporating broader Indonesian representation, multiple TCF7L2 variants and ancestry-informed analyses.

Authors' contributions

PS helped develop the study concept and methods, conduct the investigation, curate and formally analyse the data, and write the first manuscript draft. MF undertook bioinformatics-related analytical work, interpreted the molecular results and critically revised the text. HS assisted with the clinical methods, interpreted the clinical results and provided critical manuscript revisions. AAZ helped perform the formal analyses, interpret the data and critically revise the text. RBL assisted with laboratory investigations, amplified DNA by PCR, validated Sanger sequencing and critically revised the text. MA and LBK helped develop the study concept and methods, supervised the work and provided critical revisions. PS and LBK attest that all raw study data are authentic. The final version was read and approved by every author.

Data availability statement

Reasonable requests for the study data may be addressed to the corresponding author, who can provide access.

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Table 1. Clinical and biochemical characteristics of study participants

Variable	Type 2 diabetes (n=111)	Controls without diabetes (n=60)	Type 1 diabetes (n=10)	P value
Age (years)	40 ± 7	32 ± 7	25 ± 6	<0.001
Sex (men/women)	23/88	4/56	2/8	0.001
BMI (kg/m²)	23.2 ± 1.4	23.6 ± 1.7	22.7 ± 1.8	0.008
Fasting glucose (mmol/L)	8.82 (7.88, 10.71)	4.83 (4.55, 5.05)	11.04 (10.55, 11.66)	<0.001
HbA1c, mmol/mol (%)	73 ± 18 (8.8 ± 1.7)	32 ± 2 (5.1 ± 0.2)	68 ± 13 (8.4 ± 1.1)	<0.001
Insulin (mIU/L)	9.8 ± 10.4	7.4 ± 1.4	2.0 ± 0.3	<0.001
C-peptide (nmol/L)	0.72 (0.53, 0.84)	0.77 (0.55, 0.84)	0.10 (0.07, 0.10)	<0.001
HOMA-IR	3.5 (2.8, 3.9)	1.5 (1.3, 1.7)	1.0 (0.9, 1.1)	<0.001

Summaries report means with standard deviations, medians with interquartile ranges, or frequencies. Column headings show enrolled group sizes; summaries use available data for each variable. Group comparisons used one-way analysis of variance for continuous measurements following a normal distribution and the Kruskal-Wallis procedure otherwise; Fisher's exact procedure assessed sex frequencies. BMI, body mass index; HbA1c, glycated haemoglobin; HOMA-IR, homeostasis model assessment of insulin resistance.

Table 2. rs12255372 genotype and allele frequencies in cases and controls and the estimated diabetes association

Measure	Type 2 diabetes (n=111)	Controls without diabetes (n=60)	OR (95% CI)	P value
GG genotype	107 (96%)	58 (97%)	Reference	—
GT genotype	4 (3.6%)	2 (3.3%)	1.08 (0.19, 6.16)	0.930
G allele	218 (98%)	118 (98%)	—	0.920
T allele	4 (1.8%)	2 (1.7%)	—	—

Entries give counts followed by percentages. Fisher's exact procedure provided the P values. Control genotypes conformed to Hardy-Weinberg expectations (P = 1.000). CI, confidence interval; OR, odds ratio.

Table 3. Genotype-specific C-peptide and HOMA-IR measurements at rs12255372

Outcome	GG (n=175)	GT (n=6)	P value
C-peptide (nmol/L)	0.70 (0.50, 0.83)	0.79 (0.70, 0.86)	0.157
HOMA-IR	2.7 (1.6, 3.6)	3.0 (1.9, 3.7)	0.609

Data are median (interquartile range). Column headings show genotype group sizes. Exploratory analyses used available data for each outcome within the genotyped cohort, including the Type 1 diabetes descriptive group. Genotype comparisons used the Mann-Whitney U procedure to obtain P values. HOMA-IR, homeostasis model assessment of insulin resistance.

Table 4. Adjusted estimates for Type 2 diabetes case status from the logistic model

Variable	Adjusted OR	95% CI	P value
rs12255372 genotype	0.646	0.070, 5.969	0.700
Age	1.323	1.199, 1.459	<0.001
BMI	0.752	0.677, 0.835	<0.001
Sex	3.077	1.020, 9.283	0.046

The primary case-control logistic model estimated adjusted ORs with genotype, age, BMI and sex entered together as predictors. Outcome coding: controls=0 and Type 2 diabetes=1. Genotype coding: GG=0 and GT=1. Sex coding: women=0 and men=1. BMI, body mass index; CI, confidence interval; OR, odds ratio.

Figure legend

Figure 1. Representative Sanger sequencing chromatograms of TCF7L2 rs12255372. (a) GG genotype, showing a single guanine peak at the polymorphic locus. (b) GT genotype, showing overlapping guanine and thymine peaks.